

Notes: Scheinkman and Xiong (JPE, 2003)

Overconfidence and Speculative Bubbles

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1 Big picture

- **Question.** Can overconfidence and short-sale constraints generate asset price bubbles, high trading volume, and excess price volatility in a unified equilibrium model?
- **Main answer.** Yes. When investors disagree because each group overweights its own signal, and when short sales are ruled out, asset ownership includes a resale option: the owner can sell later to investors who become more optimistic.
- **Core mechanism.** A buyer is willing to pay more than his own discounted value of dividends because he also buys the option to resell the asset to another group at an even higher valuation in the future.
- **Bubble definition.** The bubble is the resale option component of price, namely the gap between the current owner's demand price and the owner's own fundamental valuation.
- **Why overconfidence matters.** Agents are not assumed to be permanently optimistic. Instead, each group overestimates the precision of its own signal. Relative optimism switches over time as signals arrive.
- **Why short-sale constraints matter.** If pessimists could short freely, they could push prices closer to fundamentals. With short-sale constraints, the asset is held by the most optimistic investors.
- **Main equilibrium object.** The equilibrium is described by a trading boundary k^* in the difference in beliefs. When the difference in beliefs crosses this boundary, the current owner sells to the other group.
- **Predictions.** Higher overconfidence raises the trading barrier, bubble size, and excess volatility; more informative private signals generally increase speculative trading opportunities; trading costs reduce turnover sharply but reduce bubble size and volatility only modestly.
- **Policy result.** A Tobin tax or transaction cost can reduce speculative trading volume substantially, but may have limited effect on the size of the bubble and on price volatility.
- **Corporate finance implication.** The resale option can violate the law of one price. In a stylized example, a subsidiary can trade for more than its parent because speculative option value is larger for the less diversified claim.

2 Incremental contribution of the paper

2.1 Contribution relative to Harrison and Kreps

- Harrison and Kreps show that heterogeneous beliefs plus short-sale constraints can generate speculative prices above fundamental values.
- Scheinkman and Xiong provide a continuous-time version with an explicit source of belief disagreement: overconfidence about signal precision.
- Their model produces an almost analytic equilibrium, making it possible to derive comparative statics for prices, trading volume, and volatility.

Interpretation: The paper turns the Harrison-Kreps insight into a tractable asset-pricing laboratory with observable predictions.

2.2 Contribution relative to behavioral finance

- The paper does not simply assume investors are optimistic.
- Investors are symmetric on average: each group overweights a different signal.
- Bubbles arise from alternating relative optimism, not from a common upward bias in beliefs.

- This structure links overconfidence to disagreement, trading volume, and volatility.

Interpretation: The paper’s behavioral assumption is disciplined: overconfidence increases disagreement and trading, while preserving symmetry between investor groups.

2.3 Contribution relative to bubble and limits-to-arbitrage models

- The model explains high prices and high turnover simultaneously.
- It shows why speculative prices can remain high even when disagreement is small: the resale option can have value because disagreement will fluctuate in the future.
- It provides a policy analysis of trading costs, showing why turnover is easier to reduce than bubbles or volatility.

Interpretation: The paper’s distinctive contribution is the joint explanation of three bubble facts: excessive valuation, high trading volume, and excess volatility.

2.4 Contribution relative to corporate finance applications

- The paper connects speculative asset pricing to real corporate phenomena such as carve-outs, spinoffs, IPO underpricing, name changes, and diversification discounts.
- It shows that option-value components can make a subsidiary appear more valuable than a parent company, as in the Palm/3Com episode.
- This suggests that managers may have incentives to create securities or corporate structures that maximize speculative resale option value.

Interpretation: The paper’s framework extends beyond secondary-market trading. It helps explain why firms may create claims that attract disagreement and speculative turnover.

3 Model setup

3.1 Dividend and fundamental processes

The risky asset pays cumulative dividends D_t :

$$dD_t = f_t dt + \sigma_D dZ_t^D. \quad (1)$$

The fundamental component f_t is latent and follows a mean-reverting process:

$$df_t = -\lambda(f_t - \bar{f})dt + \sigma_f dZ_t^f. \quad (2)$$

- f_t determines expected future dividends.
- Dividend noise prevents investors from perfectly inferring f_t .
- The risky asset is in fixed unit supply.
- Investors are risk neutral and discount at rate r .

Interpretation: Risk neutrality isolates the speculative resale channel. Dividend risk matters primarily because it makes fundamentals hard to infer.

3.2 Signals and overconfidence

All investors observe two signal processes:

$$ds_t^A = f_t dt + \sigma_s dZ_t^A, \quad (3)$$

$$ds_t^B = f_t dt + \sigma_s dZ_t^B. \quad (4)$$

There are two investor groups. Group A treats signal s^A as its own signal, and group B treats signal s^B as its own signal. Each group overestimates the informativeness of its own signal by believing its own signal innovation is correlated with the fundamental innovation:

$$ds_t^A = f_t dt + \sigma_s \phi dZ_t^f + \sigma_s \sqrt{1 - \phi^2} dZ_t^A \quad \text{in group } A\text{'s mind}, \quad (5)$$

$$ds_t^B = f_t dt + \sigma_s \phi dZ_t^f + \sigma_s \sqrt{1 - \phi^2} dZ_t^B \quad \text{in group } B\text{'s mind}. \quad (6)$$

- ϕ is the overconfidence parameter.
- Higher ϕ makes each group think its own signal is more informative.
- Agents agree to disagree about the correct interpretation of signals.

Interpretation: Overconfidence does not mean that one side is always bullish. It means each side overreacts to its own signal, causing time-varying disagreement.

3.3 Belief dynamics

Let $\hat{f}_t^A$ and $\hat{f}_t^B$ be the two groups' estimates of the current fundamental. The difference in beliefs is

$$g_t^A = \hat{f}_t^B - \hat{f}_t^A, \quad (7)$$

where g_t^A is the optimism of group B relative to group A from group A 's perspective. Proposition 1 implies that this belief difference follows an Ornstein-Uhlenbeck diffusion:

$$dg_t^A = -\rho g_t^A dt + \sigma_g dZ_t^g. \quad (8)$$

Interpretation: The stationary mean-reverting belief gap is the state variable driving speculative trades.

3.4 Original model equations panel

Read:

- The original panel shows the latent fundamental process and signal structure.
- It also shows how overconfidence is modeled as perceived correlation between own signal innovations and fundamental innovations.

Economic message: The modelling choice transforms overconfidence into a tractable Gaussian filtering problem.

parameter. The fundamental variable f is not observable. However, it satisfies

$$df_t = -\lambda(f_t - \bar{f})dt + \sigma_f dZ_t^f \quad (2)$$

where $\lambda \geq 0$ is the mean reversion parameter, $\bar{f}$ is the long-run mean of f , $\sigma_f > 0$ is a constant volatility parameter, and Z^f is a standard Brownian motion. The asset is in finite supply, and we normalize the total supply to unity.

There are two sets of risk-neutral agents. The assumption of risk neutrality not only simplifies many calculations but also serves to highlight the role of information in the model. Since our agents are risk-neutral, the dividend noise in equation (1) has no direct impact in the valuation of the asset. However, the presence of dividend noise makes it impossible to infer f perfectly from observations of the cumulative dividend process. Agents use the observations of D and any other signals that are correlated with f to infer current f and to value the asset. In addition to the cumulative dividend process, all agents observe a vector of signals s^A and s^B that satisfy

$$ds_t^A = f_t dt + \sigma_s dZ_t^A \quad (3)$$

and

$$ds_t^B = f_t dt + \sigma_s dZ_t^B, \quad (4)$$

where Z^A and Z^B are standard Brownian motions, and $\sigma_s > 0$ is the common volatility of the signals. We assume that all four processes Z^D , Z^f , Z^A , and Z^B are mutually independent.

Agents in group A (B) think of s^A (s^B) as their own signal, although they can also observe s^B (s^A). Heterogeneous beliefs arise because each agent believes that the informativeness of his own signal is larger than its true informativeness. Agents of group A (B) believe that innovations dZ^A (dZ^B) in the signal s^A (s^B) are correlated with the innovations dZ^f in the fundamental process, with ϕ ($0 < \phi < 1$) as the correlation parameter. Specifically, agents in group A believe that the process for s^A is

$$ds_t^A = f_t dt + \sigma_s \phi dZ_t^f + \sigma_s \sqrt{1 - \phi^2} dZ_t^A.$$

Although agents in group A perceive the correct unconditional volatility of the signal s^A , the correlation that they attribute to innovations causes them to overreact to signal s^A . Similarly, agents in group B believe that the process for s^B is

$$ds_t^B = f_t dt + \sigma_s \phi dZ_t^f + \sigma_s \sqrt{1 - \phi^2} dZ_t^B.$$

On the other hand, agents in group A (B) believe (correctly) that in-

Original model setup equations and signal interpretation.

4 Trading and equilibrium

4.1 Resale option value

Let $o \in \{A, B\}$ be the current owner's group and $\bar{o}$ the other group. The current owner chooses when to sell:

$$p_t^o = \sup_{\tau \geq 0} E_t^o \left[\int_t^{t+\tau} e^{-r(s-t)} dD_s + e^{-r\tau} (p_{t+\tau}^{\bar{o}} - c) \right]. \quad (9)$$

The equilibrium demand price is conjectured to have the form

$$p_t^o = \frac{\bar{f}}{r} + \frac{\hat{f}_t^o - \bar{f}}{r + \lambda} + q(g_t^o). \quad (10)$$

- The first term is the present value of long-run dividends.
- The second term is the owner's current assessment of future dividend fundamentals.
- The third term $q(g_t^o)$ is the resale option value.

Interpretation: The bubble is not imposed exogenously. It is the endogenous value of the option to resell to a future more optimistic buyer.

4.2 Trading boundary

The owner sells when the other group becomes sufficiently more optimistic. The equilibrium is characterized by a trading boundary k^* .

- If g_t^o is below the boundary, the owner keeps the asset.
- If g_t^o hits k^* , the owner sells and ownership switches.
- When trading costs are zero, the boundary tends toward zero: the owner sells as soon as the other group is even slightly more optimistic.

The bubble at the trading point is

$$b = q(-k^*). \quad (11)$$

Interpretation: The trading rule is a continuous-time optimal stopping problem. The value function is recursive because the new owner also receives a resale option.

4.3 Equilibrium theorem panel

Read:

- The original panel states the existence of a unique trading threshold for each trading cost.
- It also states that the constructed option value is an equilibrium option value.
- The panel shows the analytic expression for the resale option value $q(x)$ and the bubble at the trading point.

Economic message: These results are what make the model more than a verbal bubble story: the resale option can be solved and compared across parameter values.

The next theorem shows that, for each c , there exists a unique pair (k^*, β_1) that solves equations (19) and (20). The smooth pasting conditions are sufficient to determine the function q and the “trading point” k^* .

THEOREM 1. For each trading cost $c \geq 0$, there exists a unique k^* that solves (20). If $c = 0$, then $k^* = 0$. If $c > 0$, $k^* > c(r + \lambda)$.

The next theorem establishes that the function q described by equation (18), with β_1 and k^* given by (19) and (20), solves (13). The proof consists of two parts. First, we show that (14) and (15) hold and that q' is bounded. We then use a standard argument to show that in fact q solves equation (13) (see, e.g., Kobila [1993] or Scheinkman and Zariphopoulou [2001] for similar arguments).

THEOREM 2. The function q constructed above is an equilibrium option value function. The optimal policy consists of exercising immediately if $g^o > k^*$; otherwise wait until the first time in which $g^o \geq k^*$.

It is a consequence of theorem 2 that the process g^o will have values in $(-\infty, k^*)$. The value k^* acts as a barrier, and when g^o reaches k^* , a trade occurs, the owner’s group switches, and the process is restarted at $-k^*$. The function $q(g^o)$ is the difference between the current owner’s demand price and his fundamental valuation and can be legitimately called a bubble. When a trade occurs, this difference is

$$b \equiv q(-k^*) = \frac{1}{r + \lambda} \frac{h(-k^*)}{h'(k^*) + h'(-k^*)}. \quad (21)$$

Using equation (21), we can write the value of the resale option as

$$q(x) = \begin{cases} \frac{b}{h(-k^*)} h(x) & \text{for } x < k^* \\ \frac{x}{r + \lambda} + \frac{b}{h(-k^*)} h(-x) - c & \text{for } x \geq k^*. \end{cases} \quad (22)$$

Original theorem and option-value expressions.

5 Equilibrium implications

5.1 Trading volume

The expected duration between trades depends on the time it takes the belief gap to reach the trading boundary. When overconfidence increases the volatility of belief differences, the boundary is hit more often unless the boundary adjusts enough to offset the volatility increase.

- High overconfidence creates more disagreement.
- More disagreement creates more opportunities for resale.
- More opportunities for resale imply higher trading volume.

Interpretation: The model explains why bubbles are often accompanied by unusually high turnover.

5.2 Bubble magnitude

The bubble component is the speculative resale option value:

$$q(g_t^o) = p_t^o - \left(\frac{\bar{f}}{r} + \frac{\hat{f}_t^o - \bar{f}}{r + \lambda} \right). \quad (12)$$

- The option is more valuable when future disagreement is more volatile.
- The option is also more valuable when trading opportunities arise more frequently.
- Even small current disagreement can support a large bubble if future belief reversals are likely.

Interpretation: This is why the model can generate large bubbles even when current differences in beliefs appear modest.

5.3 Excess volatility

The asset price contains both a fundamental component and the resale option component. Fluctuations in $q(g_t^o)$ produce an extra volatility component beyond fundamental volatility.

$$\text{Vol}(p_t) = \text{Vol}(\text{fundamental value}) + \text{Vol}(\text{resale option value}) + \text{covariance terms.} \quad (13)$$

Interpretation: Volatility is not simply dividend risk. In a speculative market, changes in the resale option can generate additional price movement.

6 Comparative statics

6.1 Overconfidence

Read:

- Figure 1 shows how equilibrium objects change with the overconfidence parameter ϕ .
- Higher ϕ raises the trading barrier k^* .
- Higher ϕ raises the bubble at the trading point.
- Higher ϕ raises the extra volatility component.
- Expected time between trades is nonmonotonic because the belief process becomes more volatile but the trading barrier also changes.

Economic message: Greater overconfidence strengthens the speculative component of prices and makes resale opportunities more valuable.

6.2 Information in signals

Read:

- Figure 2 varies the informativeness of the signals.
- More informative signals reduce the trading barrier and the expected duration between trades.
- The bubble at the trading point rises.
- The extra volatility component is relatively flat in the calibration shown.

Economic message: Better signals can intensify speculation because they create sharper swings in relative beliefs.

6.3 Figures 1 and 2

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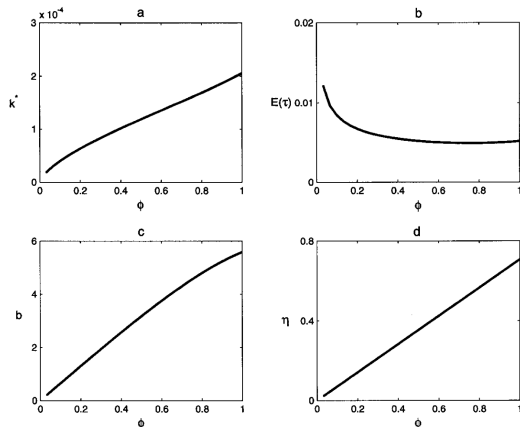


FIG. 1.—Effect of overconfidence level: *a*, trading barrier; *b*, duration between trades; *c*, bubble; *d*, extra volatility component. Here, $r = 5$ percent, $\lambda = 0$, $\theta = 0.1$, $i_s = 2.0$, $i_b = 0$, and $c = 10^{-6}$. The values of the bubble and the extra volatility component are computed at the trading point. The trading barrier, the bubble, and the extra volatility component are measured as multiples of $\sigma/(r + \lambda)$, the fundamental volatility of the asset.

Figure 1: effect of overconfidence.

6.4 Trading costs

Read:

- Figure 3 varies the trading cost c .
- Trading costs increase the trading barrier k^* and sharply increase the duration between trades.
- Trading costs reduce the bubble and the extra volatility component, but the effect on these objects is limited relative to the effect on turnover.
- Proposition 4 formalizes the result: increasing c raises k^* and reduces $q(x)$ and the extra volatility component.

Economic message: Transaction taxes are much more effective at reducing trading volume than at eliminating bubbles or speculative volatility.

OVERCONFIDENCE

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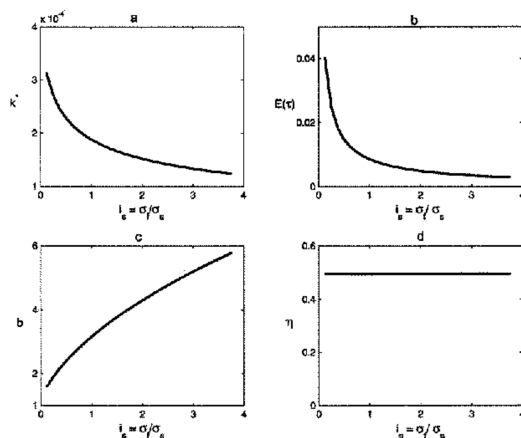


FIG. 2.—Effect of information in signals: *a*, trading barrier; *b*, duration between trades; *c*, bubble; *d*, extra volatility component. Here, $r = 5$ percent, $\lambda = 0$, $\theta = 0.1$, $\phi = 0.7$, $i_b = 0$, and $c = 10^{-6}$. The values of the bubble and the extra volatility component are computed at the trading point. The trading barrier, the bubble, and the extra volatility component are measured as multiples of $\sigma/(r + \lambda)$, the fundamental volatility of the asset.

Figure 2: effect of information in signals.

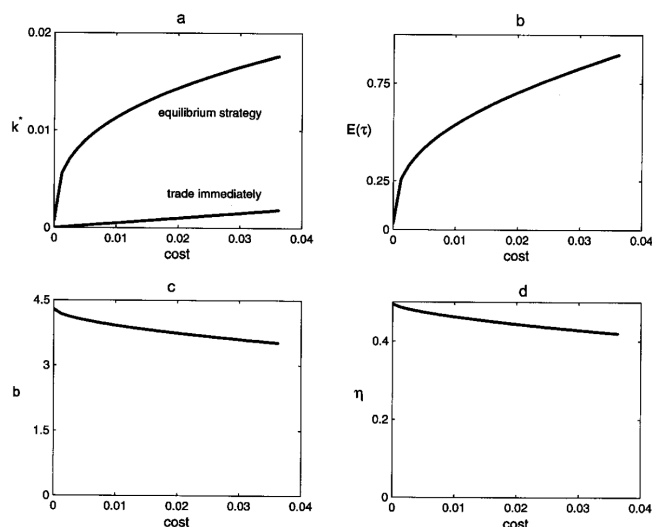


FIG. 3.—Effect of trading costs: *a*, trading barrier; *b*, duration between trades; *c*, bubble; *d*, extra volatility component. Here, $r = 5$ percent, $\phi = 0.7$, $\lambda = 0$, $\theta = 0.1$, $i_i = 2.0$, and $i_b = 0$. The values of the bubble and the extra volatility component are computed at the trading point. The trading barrier, the bubble, the extra volatility component and trading

Figure 3: effect of trading costs.

7 Policy implications and corporate finance implications

7.1 Tobin tax and transaction costs

- A small tax raises the cost of flipping the asset to a future optimist.
- This increases the threshold needed to make resale attractive.
- As a result, volume can fall sharply even if the bubble remains sizable.
- The reason is that less frequent trades can be offset by higher expected speculative profit per trade.

Interpretation: Policy can dampen speculative turnover without fully removing speculative prices.

7.2 Subsidiary-parent valuation violations

The model can generate a violation of the law of one price. Suppose firms 1 and 2 are subsidiaries of firm 3, and firm 3's dividend is the sum of the dividends of firms 1 and 2. If disagreement about each subsidiary is large, the resale option value embedded in one subsidiary can exceed the value of the diversified parent.

- The parent has less fundamental uncertainty because diversification offsets shocks.
- The subsidiary has more disagreement and hence a larger resale option component.
- The subsidiary can trade above the value of the parent stake, as in the Palm/3Com episode.

Interpretation: The model links speculative option value to apparent arbitrage violations in carve-outs and spinoffs.

7.3 Diversification discount

The model also suggests a reason diversified firms may trade at a discount relative to focused firms. Diversification reduces fundamental uncertainty and hence reduces disagreement and speculative option value.

Interpretation: A lower speculative component can make a diversified firm appear undervalued relative to a portfolio of speculative focused claims.

8 Identification and interpretation

8.1 What the model identifies theoretically

- The model identifies a mechanism by which belief disagreement creates prices above fundamentals.
- It links short-sale constraints to ownership by optimists.
- It shows how option value creates trading volume and excess volatility.
- It provides comparative statics in overconfidence, signal precision, interest rates, and trading costs.

Interpretation: This is a theoretical identification exercise: the model clarifies which assumptions generate which bubble facts.

8.2 Empirical predictions

The model predicts that speculative episodes should feature:

- high valuations relative to fundamentals;
- high turnover;
- high volatility;
- stronger effects where short-sale constraints are more binding;
- reductions in turnover after transaction costs rise;
- weaker reductions in bubble size and volatility after the same cost shock;
- larger speculative components for claims with more uncertainty and disagreement.

Interpretation: The model is useful because it makes joint predictions about prices, volume, and volatility, rather than treating bubbles as only high prices.

8.3 Limits of the framework

- Investors are risk neutral, so the model abstracts from risk premia.
- Short sales are ruled out rather than made costly in a richer security-lending market.
- Belief disagreement comes from a particular form of overconfidence.
- The model is highly stylized and does not estimate parameters on bubble episodes.

Interpretation: These simplifications create tractability. The payoff is a clear equilibrium link between overconfidence, speculative option value, turnover, and volatility.

9 Short summary

- Overconfidence makes each investor group overweight its own signal.
- Belief differences fluctuate over time as signals arrive.
- Short-sale constraints place the asset in the hands of optimists.
- Ownership gives a resale option: the owner can sell later to another group when that group becomes more optimistic.
- The asset price equals the owner's fundamental valuation plus the resale option value.
- The resale option is the bubble component.
- A trading threshold determines when the owner sells.
- Higher overconfidence raises bubble size and excess volatility.
- Trading costs sharply reduce turnover but reduce bubbles and volatility only modestly.
- The model explains why bubbles are associated with high prices, high volume, and high volatility.
- It also explains why subsidiary prices can exceed parent-company values when speculative option value is concentrated in the subsidiary.

10 Conclusion

10.1 Core conclusion

Scheinkman and Xiong provide a tractable model in which overconfidence creates disagreement, short-sale constraints prevent pessimists from correcting prices, and asset ownership contains a resale option. This option is the bubble component of price.

10.2 Economic intuition

The key intuition is that buyers do not buy only dividends. They also buy the right to sell later to someone more optimistic. When beliefs fluctuate and short sales are limited, this right can be valuable. The speculative value of future resale can create a price above the buyer's own estimate of fundamentals.

10.3 Policy implication

Transaction costs can reduce speculative volume because they make resale less attractive. But they do not eliminate the resale option quickly, because fewer trades may be offset by larger speculative profits per trade. Hence a Tobin tax may reduce turnover more than it reduces bubbles.

10.4 Takeaway

The paper's central lesson is that bubbles are not just high prices. In markets with disagreement and short-sale constraints, bubbles naturally come with high turnover and high volatility. Any convincing theory of bubbles should account for all three facts jointly.